

Sales Prediction using Python

Objective

The objective of this project is to predict product sales based on advertising expenditures across different media channels.

Dataset

The dataset contains:

- TV Advertising Budget
- Radio Advertising Budget
- Newspaper Advertising Budget
- Sales

Technologies Used

- Python
- Pandas
- Scikit-learn
- Matplotlib
- Linear Regression

Methodology

1. Load and preprocess the dataset.
2. Select advertising features as input variables.
3. Train a Linear Regression model.
4. Predict future sales.
5. Evaluate performance using R^2 Score and MSE.
6. Analyze the impact of advertising channels on sales.

Results

The model achieved:

- R^2 Score = 0.8994
- MSE = 3.1741

Radio advertising showed the strongest influence on sales, followed by TV advertising.

Conclusion

This project demonstrates the practical application of machine learning in sales forecasting and marketing strategy optimization.